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Physiologically Based Pharmacokinetic Model of Brigatinib in Healthy Volunteers and Patients With Cancer

Yichao Xu  | Xinhua Hu  | Pengfei Zhao  | Lu Wang  | Zourong Ruan  | Bo Jiang  | Honggang Lou 

Center of Clinical Pharmacology, Second Affiliated Hospital of Zhejiang University, Hangzhou, China

Correspondence: Honggang Lou (louhg@zju.edu.cn)**Received:** 30 October 2025 | **Revised:** 6 July 2026 | **Accepted:** 10 July 2026**Keywords:** brigatinib | Chinese population | drug–drug interactions | hepatic or renal impairment | physiologically based pharmacokinetic model

ABSTRACT

Brigatinib, an oral ALK inhibitor for metastatic NSCLC, lacks dosing guidance for special populations such as the Chinese. This study developed a physiologically based pharmacokinetic (PBPK) model using European data from patients with hepatic/renal impairment and drug–drug interaction (DDI) studies (itraconazole, rifampin). The model was then applied to predict (1) pharmacokinetics (PK) in the Chinese population; (2) PK in Chinese patients with hepatic/renal impairment; and (3) DDI in Chinese patients. Validated against clinical data, the model successfully predicted brigatinib PK alone and with CYP3A4 modulators. Food simulations showed a slight absorption delay without clinically meaningful exposure reduction. In hepatic/renal impairment, the model accurately predicted exposure changes across severity groups (fold error < 2). Extending the European-validated model to Chinese populations, the findings demonstrate reliable predictions of brigatinib PK in Chinese individuals as well as in Chinese patients with hepatic or renal impairment.

1 | Introduction

Cancer is a major public health issue of the 21st century, affecting millions of patients and their families worldwide every day. However, we are not entirely hopeless in the fight against cancer. Brigatinib, a novel anticancer agent, is bringing new hope to many cancer patients [1]. Brigatinib is an oral, selective inhibitor of anaplastic lymphoma kinase (ALK), a common genetic mutation found in various tumors. By inhibiting ALK activity, brigatinib prevents cancer cell growth and spread. Studies have demonstrated its significant therapeutic effects in ALK-positive non-small cell lung cancer (NSCLC) patients [2–4]. In 2017, the U.S. Food and Drug Administration (FDA) approved brigatinib for the treatment of ALK-positive NSCLC. Although brigatinib has shown promising efficacy, like other anticancer drugs, it can cause side effects. Common adverse effects include rash, diarrhea, nausea, and vomiting [5, 6]. However, most of these reactions are manageable, and patients can continue treatment with dose adjustments or symptomatic support [7].

Throughout the history of drug use, medication management for special populations has remained a challenge. Their unique physiological characteristics, such as hepatic or renal impairment, lead to significantly different drug responses compared to the general population. A dedicated study in patients with severe renal impairment supports a 50% dose reduction of brigatinib [8], as the unbound area under the concentration–time curve from time 0 to infinity ($AUC_{0-\infty}$) in these patients is approximately twice that of patients with normal renal function. Another study in patients with hepatic impairment supports no dose adjustment for mild to moderate impairment (Child-Pugh A and B), but recommends a 40% dose reduction for severe impairment (Child-Pugh C), because the systemic exposure of unbound brigatinib ($AUC_{0-\infty}$) is increased by approximately 37% in these patients [9].

Brigatinib has an oral bioavailability of approximately 90% (FDA review document), indicating nearly complete absorption. Less than 5% of the dose is excreted unchanged in urine, suggesting a

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Study Highlights

- What is the current knowledge on the topic?
 - Brigatinib is a novel oral tyrosine kinase inhibitor approved in several countries for the treatment of patients with metastatic non-small cell lung cancer who are kinase-positive in anaplastic lymphoma. There are few reports on the optimal dose of brigatinib to be administered in special populations and in the Chinese population, and it remains a challenge to maximize the benefit from anticancer therapy by enabling patients to tolerate the highest therapeutic dose for the longest period of time.
- What question did this study address?
 - A physiologically based pharmacokinetic (PBPK) model was developed to characterize the pharmacokinetics of brigatinib in European and Chinese populations and to predict the pharmacokinetics in healthy populations and in patients with impaired hepatic and renal function. This study also investigated the effects of itraconazole and rifampicin on brigatinib pharmacokinetics (PK) to explore drug–drug interactions.
- What does this study add to our knowledge?
 - The PBPK model developed in this study successfully predicted the PK profile of brigatinib alone and in interaction with CYP3A4 inducers or inhibitors. In patients with impaired hepatic and renal function, the PBPK model accurately predicted the changes in brigatinib exposure in different impaired function groups. Meanwhile, we extended the PBPK model, which had been validated in European populations, to simulate brigatinib exposure in different Chinese populations.
- How might this change drug discovery, development, and/or therapeutics?
 - The PBPK model established in this study helps to more accurately describe the PKs of brigatinib in European and Chinese populations and patients with hepatic and renal impairment and further explores drug–drug interactions mediated by the metabolic enzyme CYP3A4.

minor role of renal clearance. In vitro studies show that CYP3A4 accounts for approximately 80%–90% of total oxidative metabolism, with CYP2C8 contributing the remainder [4, 10].

The emergence of physiologically based pharmacokinetic (PBPK) modeling provides a new solution [11–13]. The PBPK model is a physiology-based mathematical approach that simulates the dynamic processes of drugs in the human body. First proposed in the 1930s, after decades of development, it has become an essential tool for drug research, development, and clinical medicine [14–17]. Due to differences in physiological functions, the response of special populations to drugs differs significantly from that of the general population, necessitating personalized medication guidance based on individual

conditions. The PBPK model makes this possible. For example, in patients with hepatic impairment, whose livers cannot metabolize drugs normally, dosage adjustments are often necessary. Using the PBPK model, we can predict the drug's dynamic behavior in the body based on the patient's specific liver function, thereby providing personalized medication advice [18, 19].

Since brigatinib tablets were approved by the Chinese National Medical Products Administration (NMPA) in March 2022, they have been indicated as a monotherapy for patients with locally advanced or metastatic ALK-positive NSCLC. To our knowledge, there is currently a significant gap in the application of brigatinib in Chinese patients, especially those with renal or hepatic impairment. Moreover, drug–drug interactions cannot be overlooked. Therefore, in this study, we plan to use a PBPK model to predict the pharmacokinetic (PK) characteristics of brigatinib in Chinese patients. The main objectives of this study are: (1) to establish an adult PBPK model for brigatinib; (2) to predict the pharmacokinetics of brigatinib in Chinese patients with hepatic impairment, renal impairment, and under CYP3A4-mediated drug–drug interactions; and (3) to propose dose adjustment recommendations for these special populations.

2 | Methods

2.1 | Ethics and Patients

In this study, data from food effect studies [2, 19–22] in healthy subjects, PK studies [8, 9] in patients with hepatic and renal impairment, and drug interaction studies [20] were retrospectively collected for the purpose of establishing and validating the PBPK model. This study was conducted in accordance with the International Conference on Harmonization Good Clinical Practice Guidelines and local regulations, and was approved by an independent review board at each participating center.

2.2 | Modeling Platform and Data Collection

The open-source PK-Sim software (Open Systems Pharmacology Suite, version 11.1) was used to establish PBPK models for brigatinib in healthy European and Chinese subjects and patients with hepatic and renal impairment. The workflow of our study is described in Figure S1. WebPlotDigitizer (version 4.2) software was used to extract PK data points from PK curves reported in the literature, and Phoenix WinNonlin (Pharsight, Mountain View, CA, USA, version 8.3.5.340) software was used for non-compartmental model analysis to calculate PK parameters. Platform Validation was performed per FDA/EMA requirements to ensure software integrity, mathematical correctness, and algorithmic accuracy. This includes verification of in silico solver performance, physiological model structure, and system-dependent parameters. Context-of-use specific qualification was conducted using clinical PK data from ≥ 3 independent studies. The model's predictive capability was confirmed via visual predictive checks and fold-error analysis. It is important to note that all simulated and observed plasma concentrations presented in this study refer

to total (bound plus unbound) drug concentrations, unless otherwise specified. This is consistent with the standard practice for clinical pharmacokinetic reporting and ensures direct comparability across all analyzed populations and scenarios.

2.3 | PBPK Model Development and Validation

In this study, a PBPK model was developed using a combination of bottom-up (physicochemical properties and in vitro parameters informed initial model structure) and middle-out approaches (leveraged structural similarities to ALK inhibitors and stepwise parameter optimization). The PBPK model was developed using 18 compartments in PK-Sim software, with each compartment further subdivided into sub-compartments. The absorption, distribution, metabolism, and excretion (ADME) mechanisms of brigatinib were elucidated by collecting database data. Model parameters were derived

exclusively from published in vitro and clinical data, with no new bench experiments performed, and extrapolating in vitro research data to the human body by integrating in vitro-related factors and scalars. Table 1 summarizes all compound input parameters used in the development and validation of the PBPK model.

Virtual populations were initialized using PK-Sim's European/Japanese templates, with key parameters refined using average values from literature. Where Chinese data were unavailable, the Japanese population served as a physiologically justified surrogate with age, weight, and body mass index (BMI) of 44/44 years, 75.0/61.9 kg, and 26.0/21.7 kg/m², respectively, to represent the European/Chinese healthy subject population. In addition, based on this virtual individual, Virtual cohorts (*n* = 100, 50% female, age 18–60 years) were designed per regulatory standards [FDA 2018] to capture population variability while excluding age extremes requiring specialized

TABLE 1 | Summary of compound parameters of brigatinib, itraconazole, and rifampin.

Parameter	Brigatinib	Source	Itraconazole	Rifampin	Source
Physiochemical properties					
log <i>p</i>	1.80	ChemAxon/ Simulated	4.62	2.50	Template in PK Sim
<i>f_u</i> (plasma, albumin)	0.66	[23]	0.006	0.17	Template in PK Sim
MW (g/mol)	584.1	PubChem	661.6	822.9	Template in PK Sim
pKa (acid)	12.88	ChemAxon	NA	7.90	Template in PK Sim
pKa (base)	8.54	ChemAxon	3.70	1.70	Template in PK Sim
Solubility (mg/mL)	0.022	ALOGPS	0.008	2.8	Template in PK Sim
Absorption	3.61E−07	PK Sim Calculation	5.33E−05	1.24E−05	Template in PK Sim
Specific intestinal					
Permeability (cm/min)					
Distribution	4.62E−05	PK Sim Calculation			
Specific organ					
permeability					
permeability (cm/min)			0.01	2.93E−05	Template in PK Sim
Metabolism and elimination					
CYP3A4					
kcat (L/min)	1.55	Fitting by FDA Reports	0.04	9.87	Template in PK Sim
Inhibition Ki (nmol/L)	NA		1.3	18,500	Template in PK Sim
Induction EC50 (μmol/L)	NA		NA	0.34	Template in PK Sim
CYP2C8					
kcat (L/min)	4.0	Fitting by FDA Reports	NA	NA	Template in PK Sim
Renal clearance (CL _R)	1.9 L/h	[24, 25]			

Note: The link to FDA reports: https://www.accessdata.fda.gov/drugsatfda_docs/nda/2017/208772Orig1s000MultidisciplineR.pdf. Abbreviations: *f_u*, fraction unbound; log *p*, lipophilicity; MW, molecular weight; PBPK, physiologically-based pharmacokinetic; pKa, acid dissociation constant.

models [26]. The virtual population was created using the software's population module to characterize the pharmacokinetic (PK) behavior of brigatinib in European and Chinese populations. Using the ACAT model with permeability-limited absorption, food effect simulations compared fasted (> 8 h) with FDA-standard high-fat. The base PBPK model was developed using the fasted condition from the food effect study [27] (single 180 mg dose) and the brigatinib control arm from the DDI study [10] (single 90 mg dose) as training datasets. The model was then validated against independent datasets, including the fed condition from the food effect study, the itraconazole and rifampin arms from the DDI study, and the hepatic impairment study [9] and renal impairment study [8], without any further parameter adjustment.

The overall predicted mean and the 5th–95th percentile concentration range obtained will be compared with the measured values to evaluate the predictive performance of the model. In addition, the predictive performance will be assessed by comparing the predicted concentrations with the measured maximum concentration ($C_{\max}$) and the area under the curve from zero to infinity ($AUC_{0-\infty}$) over the average fold error (AFE) to evaluate the fit of the model.

2.4 | Parameter Sensitivity Analysis

Before PBPK prediction, it is critical to understand how parameter uncertainty affects the model's simulation results. Using existing databases and literature reports, a variety of data and information sources were used to optimize the PBPK model parameters and perform a parameter sensitivity analysis. To maintain numerical stability, sensitivity calculations were based on the average of several sensitivities for different changes. To determine relative changes, the simulated values were multiplied by one of the following change factors: 1.10, 1.05, 1/1.05, and 1/1.10. The sensitivity of PK parameters to input parameters was then calculated by dividing the relative change in PK parameters by the relative change in input parameters. The effect of parameter changes on $AUC_{0-\infty}$ and $C_{\max}$ was used to calculate parameter sensitivity, and the results of the parameter sensitivity analysis are provided in the [Supporting Information Files](#).

2.5 | Extrapolation to the Renal Impairment Population

Based on the PBPK model for healthy individuals, and taking into account the exposures of systemic anatomy and physiology, we continue to use it to assess renal impairment. We replace the healthy virtual individuals with those adapted to renal impairment while keeping all drug-specific parameters unchanged. We then developed patients with varying degrees of renal impairment using the Chronic Kidney Disease (CKD) module built into PK-Sim. Individuals with renal impairment exhibit changes in parameters including GFR, renal volume, renal blood flow, plasma protein binding, hematocrit, gastric emptying time, and small intestine transit time [20]. Finally, we predict the renal clearance rate of European patients with renal impairment based on fGFR and creatinine clearance

rate (CL_{cr}) and validate the PBPK model based on clinical PK studies reported by Gupta et al. [8].

2.6 | Extrapolation to the Hepatic Impairment Population

For individuals with hepatic impairment, the Child-Pugh classification is the most widely used method for categorizing liver function. Based on clinical characteristics and laboratory parameter values, liver function impairment is classified into Child-Pugh Class A (CP-A), Child-Pugh Class B (CP-B), and Child-Pugh Class C (CP-C) groups [21]. We referred to the reported pathophysiological changes at different stages of hepatic impairment in the population and incorporated them into the developed PBPK model. A literature search was conducted to determine the physiological changes associated with Child-Pugh class A to C liver function impairment. Parameters included organ blood flow, plasma protein concentration, hematocrit, and liver function volume [22]. At the same time, even though the liver is not the primary route of excretion, liver disease can alter renal function and lead to the accumulation of drugs and their metabolites in the body. Therefore, based on existing reports, we also included GFR as a variable affecting pharmacokinetics in patients with hepatic impairment, and the parameter information is shown in Table S3. We predict and validate the PBPK model based on real PK studies reported by Hanley et al. [9].

2.7 | DDIs Model Development

Since brigatinib is primarily metabolized by CYP3A4 in vitro, we constructed a PBPK model to evaluate the effect of the strong CYP3A4 inhibitor itraconazole and the strong CYP3A4 inducer rifampin (Table 1) on the pharmacokinetic characteristics of brigatinib and to guide concomitant medication during brigatinib administration, and itraconazole or rifampin models were used without modification from PK-Sim. Based on the clinical PK studies reported by Tugnait et al. [10], we predict and validate the PBPK model to investigate potential drug–drug interactions between brigatinib (90/180 mg) and itraconazole (200 mg) or rifampin (600 mg).

2.8 | PBPK Model in Chinese Population and Dose Optimization

By changing the race setting from “European” to “Chinese” in the PK Sim software, simulations targeting Chinese patients were conducted using PK-Sim's Japanese population template, retaining its inherent physiological parameters, we examined the simulated exposure of brigatinib in healthy Chinese adults and evaluated dosing regimens in different Chinese populations. Specifically, by generating and combining different virtual Chinese populations, we simulated the $AUC_{0-\infty}$ values for each individual while selecting the optimal clinical dose to achieve normalization. We then compared the $AUC_{0-\infty}$ differences between each group and healthy Chinese subjects and annotated the results accordingly. To align the exposure of brigatinib with that of healthy Chinese adults, we adjusted the concentrations

up or down based on the normalization results for doses in special Chinese populations and healthy Chinese adults. Finally, we provided dosing recommendations for each subpopulation within the Chinese population.

3 | Results

3.1 | Parameter Sensitivity Analysis

Sensitivity parameter analysis showed that lipophilicity had the greatest impact on $C_{\max}$ and AUC. Gastric emptying time was identified as an important parameter that affected the $t_{\max}$ of brigatinib.

3.2 | Development and Validation of Base PBPK Models

We developed an oral PBPK model tailored to healthy European adult volunteers following the outlined development strategy. By generating simulated data, we obtained blood concentration-time profiles for healthy populations, which we then compared with actual observed data [10, 27]. As shown in Figure 1, the discrepancies between the majority of the observed oral dosing data and the predicted concentrations are limited to a range of 0.8 to 1.25 times, with all oral dosing observations falling within a twofold margin of error. The plasma concentration-time profiles are shown in Figure S2. We then further investigated the effect of food on the pharmacokinetics of brigatinib. Using PK-Sim software, we created a simulated high-fat meal scenario, with meal composition details detailed in Table S4. Based on our validated PBPK model for healthy European subjects, we included an additional event representing the consumption of a high-fat meal. By contrasting the observed data with the actual measured values, we assessed the effect of the meal on the pharmacokinetics of brigatinib, as shown in Figure 2. The PBPK

model constructed for a virtual adult population was robustly validated and is ready for use in future research efforts.

3.3 | PBPK Model in Hepatic or Renal Impairment Population

Based on PK study data from patients with varying degrees of hepatic and renal impairment receiving brigatinib, this analysis aims to explore the PK characteristics of brigatinib in populations with such impairments. Results from the extrapolation model indicate that the predicted plasma drug concentration-time profiles and actual iron levels for individuals with CP-A to CP-C hepatic impairment and those with severe renal impairment closely match the observed data (Figure 3, arithmetic mean). The majority of the observed values fall within the 5th–95th percentile range, the Geometric Mean Fold Error (GMFE) of $C_{\max}$ range from 1.18 to 1.48 and the GMFE of $AUC_{0-\infty}$ range from 1.24–1.53. The results demonstrated that the geometric mean $AUC_{0-\infty}$ in CP-A, CP-B, and CP-C hepatic impairment was approximately 89.3%, 99.2%, and 137.4% of the value observed in the healthy population, respectively. Similarly, the $AUC_{0-\infty}$ in severe renal impairment groups was approximately 177.2% of the value observed in the healthy population.

3.4 | PBPK Model for DDIs

The PBPK model was used to simulate the changes in the plasma concentration-time curves of brigatinib when used in combination with itraconazole and rifampin (Figure 4, arithmetic mean). To predict the DDI of itraconazole, healthy volunteers were administered a single oral dose of 90 mg brigatinib with or without 200 mg itraconazole; to predict the DDI of rifampin, healthy volunteers were administered a single oral dose of 180 mg brigatinib with or without 600 mg rifampin. The predicted oral PK profiles from the model were

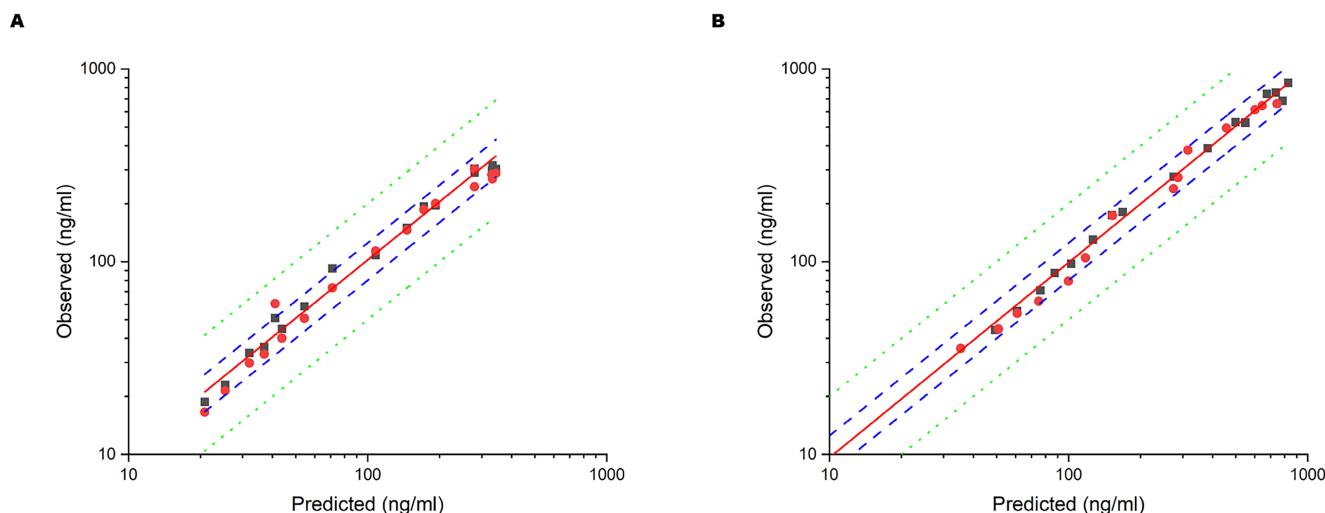


FIGURE 1 | The goodness of fit plot for brigatinib. The red solid line represents the unity line; the blue and green dashed lines indicate, respectively, 0.8–1.25 and 0.5–2 folds above and below the line of unity. (A) The red circle data is from Group A of the DDIs study by Tugnait et al. [10], and the black squares are from Group B of the DDI study by Tugnait et al. [10]; (B) the red circle data is from Group C of the DDIs study by Tugnait et al. [10], and the black squares are from the food effect study by Tugnait et al. [27].

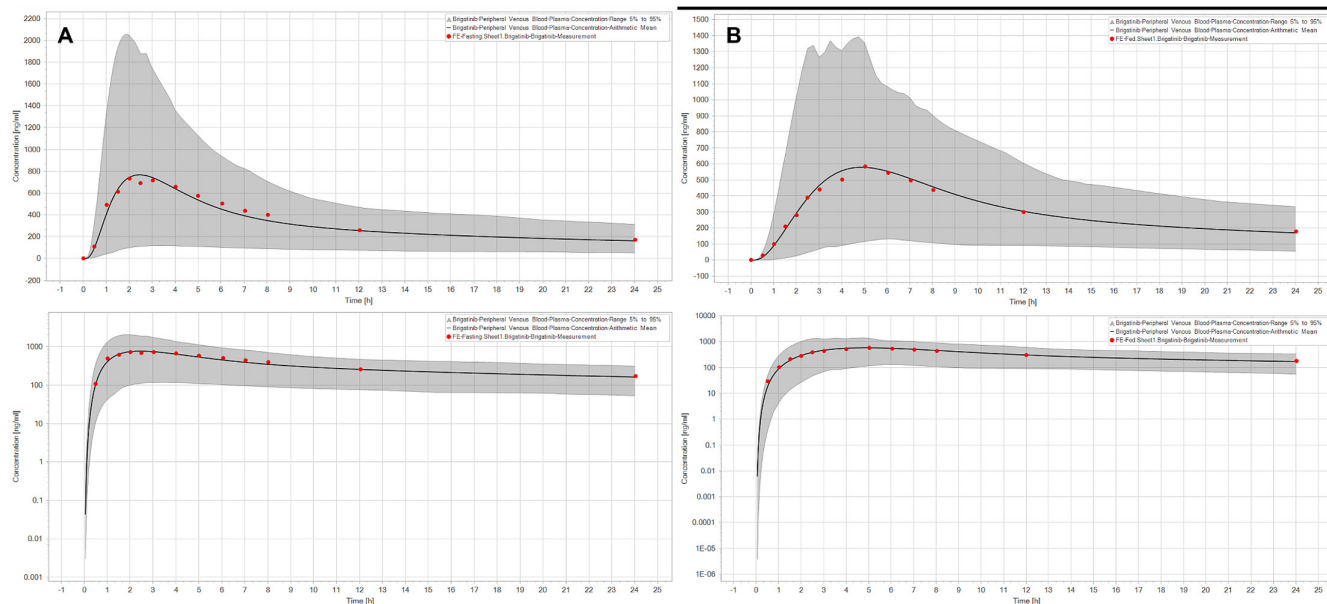


FIGURE 2 | The PBPK model simulates the food effect of brigatinib. (A) Predicted and observed brigatinib plasma concentration-time profiles after oral administration of 180mg brigatinib under fasted conditions. (B) Predicted and observed brigatinib plasma concentration-time profiles after oral administration of 180mg brigatinib under fed conditions.

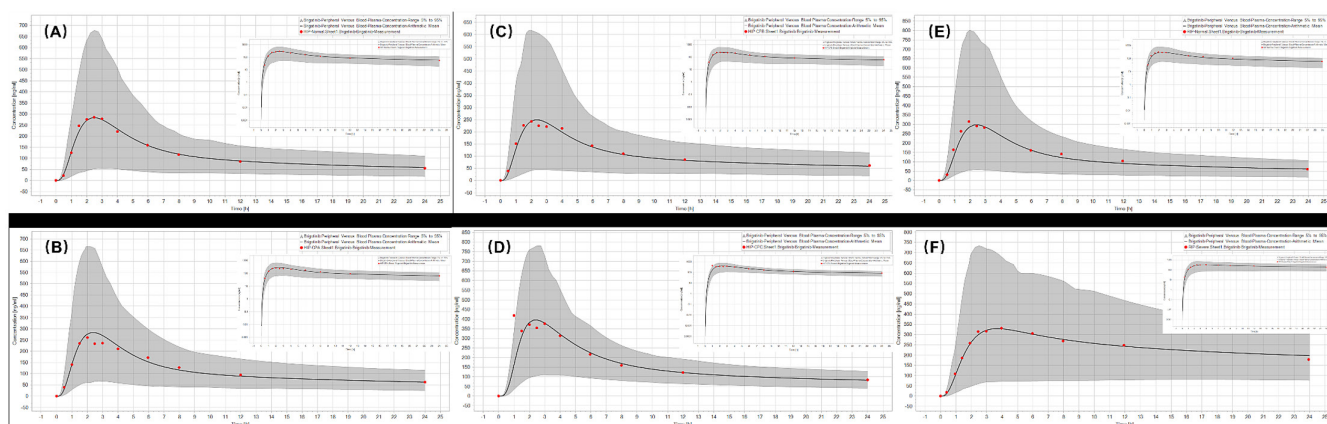


FIGURE 3 | Prediction of the pharmacokinetic profiles for brigatinib in patients with hepatic or renal impairment. (A) Normal hepatic function, (B) Child-Pugh A hepatic impairment, (C) Child-Pugh B hepatic impairment, (D) Child-Pugh C hepatic impairment, (E) Normal renal function, (F) Severe renal impairment.

consistent with the observed data, the GMFE of C_{max} ranged from 1.21 to 1.32 and the GMFE of $AUC_{0-\infty}$ ranged from 1.28 to 1.41. Although the model slightly overpredicted exposures (GMFE up to 1.41), all predictions fell within the 0.5- to 2-fold acceptance range, and the concentration-time profiles visually aligned well with observed data, confirming the model's predictive utility for DDI assessment. Itraconazole coadministration doubled the geometric mean $AUC_{0-\infty}$ of brigatinib, while rifampin coadministration reduced exposure to 20% of control.

3.5 | PBPK Model in Chinese Population

The PBPK model was used to predict the concentration-time profile of a single dose of brigatinib in different populations

and under different scenarios. We conducted separate simulations to evaluate the PK characteristics in Chinese healthy volunteers, patients with normal hepatic and renal function, and patients with hepatic or renal impairment following a 90 mg oral dose of brigatinib. We also evaluated the food effect and DDIs. The results are visualized in box plots as shown in Figure 5. A comprehensive summary of the simulated PK parameters for the Chinese population is presented in Table 2. To maintain equivalent therapeutic exposure across patient populations, dose adjustments are recommended as follows: for renal impairment, adjusted doses to approximately 130%, 122%, and 33% of healthy adult levels in mild, moderate, and severe chronic kidney disease (CKD) respectively; for hepatic impairment, reduce doses to 83% (Child-Pugh A), 76% (CP-B), and 63% (CP-C) of standard values; and during concomitant therapy, decrease doses to 75% with itraconazole 200 mg

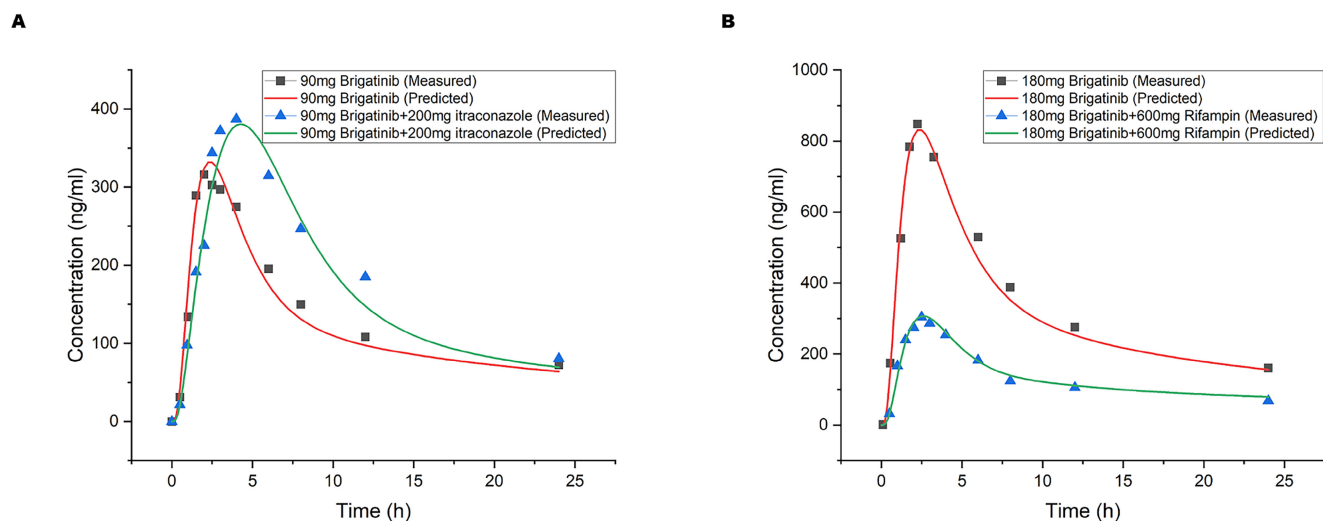


FIGURE 4 | Prediction of brigatinib's drug–drug interaction. (A) Prediction of the drug–drug interaction between intravenous brigatinib and itraconazole. (B) Prediction of the drug–drug interaction between oral brigatinib and rifampin.

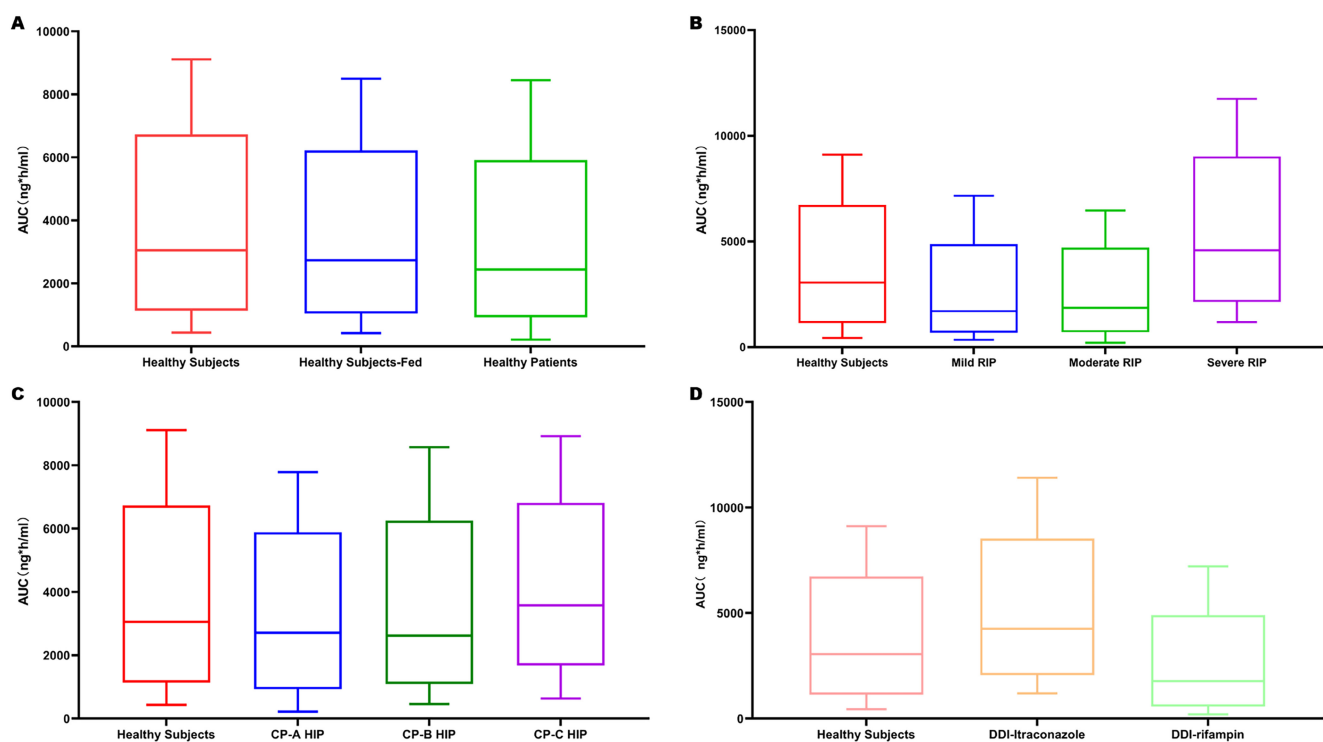


FIGURE 5 | Predicted brigatinib plasma concentration profiles in Chinese populations. (A) Box-whisker plots of healthy subjects, patient populations, and the impact of food. (B) Box-whisker plots of populations with varying stages of renal disease. (C) Box-whisker plots of populations with different stages of hepatic disease. (D) Box-whisker plots of populations experiencing drug–drug interactions. CP-A, Child-Pugh A; CP-B, Child-Pugh B; CP-C, Child-Pugh C; Healthy patients, cancer patients with normal hepatic and renal function; HIP, hepatic impairment population; RIP, renal impairment population.

(CYP3A4 inhibition) while increasing to 135% with rifampin 600 mg (CYP3A4 induction).

4 | Discussion

Lung cancer is the leading cause of cancer-related deaths in China. Today, patients with advanced NSCLC harboring positive driver genes have entered the era of targeted therapy.

Brigatinib has emerged as a prominent investigational drug in recent years. It is designed to treat patients who have developed resistance to first-generation ALK TKIs and is expected to enter the Chinese market in 2022 [28, 29].

Model-based approaches drive patient-centric and efficient drug development and are critical to the success of accelerated oncology drug development programs [30–32]. Compared to traditional PK methods, PBPK models primarily use modeling and

TABLE 2 | The mean predicted exposure of brigatinib in Chinese populations.

Parameter	Healthy subjects		Patients								
	Control group	High-fat meal	Healthy	Mild RIP	Moderate RIP	Severe RIP	CP-A HIP	CP-B HIP	CP-C HIP ^a	With 200 mg itraconazole	With 600 mg rifampin
$C_{\max}$ (ng/mL)	343	235	302	211	271	359	318	315	409	433	187
$AUC_{0-\infty}$ (ng/mL·h)	5743	5392	4991	3841	4077	15,351	5996	6537	7904	6681	3705

Note: Values are arithmetic means; medians and percentiles are provided in Supporting [Excel File S1](#).

Abbreviations: CP-A, Child-Pugh A; CP-B, Child-Pugh B; CP-C, Child-Pugh C; HIP, hepatic impairment population; RIP, renal impairment population.

^aPredicted values for mild and moderate renal impairment (CKD stages 1–3) are model-based simulations only and lack clinical validation. Only severe renal impairment predictions (CKD stage 4–5) are supported by clinical data [8]. These mild/moderate predictions should be interpreted with caution.

simulation to predict plasma drug concentrations. This approach provides a basis for accurate efficacy and safety assessments, as well as guidance for clinical drug dosing regimens in special populations [24, 25]. This study uses a PBPK-based approach spanning from healthy European and Chinese populations to patients with impaired hepatic and renal function. It evaluates potential drug–drug interactions and predicts the pharmacokinetic profile of brigatinib. Based on PK data reported in the literature, overall clearance rates in Child-Pugh Class A and B patients were not significantly different from those in healthy subjects. While decreased hepatic metabolism typically increases exposure, our simulations show lower AUC in Child-Pugh A and B due to counteracting distribution changes: a 23% increase in volume of distribution (Vd) resulting from reduced plasma protein binding and portal hypertension effects (reduced first-pass extraction partially offset by oral bioavailability from gut edema). Our PBPK simulations revealed that mild to moderate hepatic impairment (Child-Pugh A and B) did not increase total brigatinib exposure compared to healthy subjects; in fact, a modest decrease was observed. This finding may appear counterintuitive given the liver's primary role in brigatinib metabolism. However, several compensatory mechanisms can offset reduced metabolic capacity in early-stage hepatic impairment. First, decreased plasma protein binding (e.g., lower albumin and α 1-acid glycoprotein levels) increases the unbound fraction of brigatinib, which can enhance Vd and thus lower total plasma concentrations without affecting unbound exposure. Second, alterations in gut wall permeability or transit time (e.g., due to portal hypertension or intestinal edema) may reduce oral bioavailability. Third, residual CYP3A4 activity in Child-Pugh A and B patients may still be sufficient to maintain near-normal clearance, as the liver retains substantial functional reserve. In contrast, severe hepatic impairment (Child-Pugh C) overwhelms these compensatory mechanisms, leading to a marked reduction in metabolic capacity and a consequent increase in total brigatinib exposure. These observations highlight the importance of considering pathophysiology-driven changes beyond simple reductions in enzyme activity when interpreting PBPK predictions for hepatically impaired populations.

Without modifying the kinetic parameters of the metabolic enzymes in the simulation of hepatic injury populations, we could adequately fit the PBPK model for patients with hepatic impairment. Consequently, in this study, the kinetic parameters of metabolic enzymes used in the PBPK model were kept consistent with those observed in healthy adult subjects.

Gupta and associates used data from 442 individuals from five clinical trials, including three phase I studies in healthy volunteers, a phase I/II study in patients with late-stage cancer, and the phase II ALTA study in patients with advanced, crizotinib-resistant, ALK-positive non-small cell lung cancer (NSCLC), to develop a population pharmacokinetic model for brigatinib. Race was one of the variables considered [10]. While the PPK analysis suggested that race did not significantly affect clinical outcomes, the research revealed that the AUCSS was 1.07 times higher in Asian populations compared to Caucasians. When additional factors such as hepatic and renal function, food effect, and concomitant medications are considered, the individualization of brigatinib dosing in Asian populations warrants more careful attention. Our simulations within a PBPK model

for Chinese populations suggest that diet and the presence of cancer disease have a minimal effect on systemic exposure to brigatinib. However, patients taking concomitant medications that either induce or inhibit CYP3A4, as well as those with severe hepatic or renal impairment, may require adjustments in brigatinib dosing during treatment to ensure safe and effective therapy.

Although the PBPK model established in this study has a stable structure (robustness of the PBPK model architecture) and excellent predictive ability, some limitations must be considered when evaluating the results. First, all the data used in this study were extracted from the published literature using software rather than being obtained directly from the investigators. Although the PK parameters calculated using these extracted data points are comparable to those reported in the literature, minor errors are unavoidable. On the other hand, brigatinib is mainly metabolized by CYP3A4 and CYP2C8. Since the PK-Sim database contains only drugs related to CYP3A4 metabolism reported in the literature, such as rifampin and itraconazole, but not gemfibrozil, which is related to CYP2C8 metabolism, the effect of CYP3A4 inducers and inhibitors on brigatinib was considered in the simulation of DDIs. In addition, the PBPK developed in this study is used to predict the dose required for different populations to achieve the same PK exposure as the standard population, rather than the same clinical efficacy. Unfortunately, there are no PK study reports for the Chinese patient population. In this case, we used the PBPK model to simulate the PK exposure of the Chinese patient population to reflect the clinical efficacy for clinical use. So far, what we have done is just a reference for the clinical use of this drug in the Chinese patient population, and we will continue to pay attention to this issue and validate our predictions in subsequent studies.

Overall, based on the results of this study, the model can be further evaluated and optimized in future studies.

5 | Conclusion

The PBPK model established in this study helps to more accurately describe the PKs of brigatinib in European and Chinese populations and patients with hepatic and renal impairment and further explores drug–drug interactions mediated by the metabolic enzyme CYP3A4. By incorporating changes in pathophysiological factors into the model and accurately extrapolating the model to these populations, this study demonstrates a method to improve the predictive ability of drug–disease models, making them a valuable resource for future clinical individualized drug dosing and evaluation.

Author Contributions

Y.X. wrote the manuscript. Y.X. and H.L. designed the research. X.H., P.Z., Z.R., L.W., and B.J. performed the research. Y.X. and H.L. analyzed the data.

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AI tools were used to improve language.

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Ethics Statement

All patients signed informed consent forms for their respective studies, which were conducted by the International Conference on Harmonization Good Clinical Practice guidelines and local regulations and approved by the independent review boards of each participating center.

Consent

All the authors consent for publication.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data and materials included in this study are available upon request by contact with the corresponding author.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** psp470302-sup-0001-Supinfo01.rar. **Data S2:** psp470302-sup-0002-TableS1.xlsx. **Table S1:** Fundamental PK properties. **Table S2:** Clinical DDI evidence. **Table S3:** Changes of PBPK parameters altered in hepatic impaired individuals. **Table S4:** Summary of high-fat meal. **Table S5:** Individual observed and predicted geometric mean of C_{max} and AUC values for each DDI study. **Figure S1:** Generic workflow for model development. **Figure S2:** The plasma concentration-time profiles behind the GoF plots. **Data S4:** psp470302-sup-0004-TableS3.xlsx.